WORKED SOLUTION

STEP III 2001, Question 4

Pure • Inverse Trigonometric Functions

STEP III 2001, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q4

The key idea

The expression $\dfrac{2x}{1+x^2}$ should ring a bell: it is the "t-formula" for $\sin 2\theta$ when $x = \tan\theta$. Substituting $x = \tan\theta$ turns $\sin^{-1}$ of the expression into $\sin^{-1}(\sin 2\theta)$, and the whole question becomes one about when $\sin^{-1}(\sin\phi)$ really does give back ϕ. The answer depends on whether ϕ lies in the principal range $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$, and that is exactly where the split at $x = \pm 1$ comes from.

Solution

Part (i)

First some basic features of $g(x) = \dfrac{2x}{1+x^2}$.

- The denominator is never zero, so g is defined and continuous everywhere.
- $g(-x) = -g(x)$, so g is odd and its graph has rotational symmetry about the origin.
- $g(0) = 0$; $g(x)$ has the same sign as x; and $g(x) \to 0$ as $x \to \pm\infty$, since the denominator grows like x^2 and the numerator only like x. So the x-axis is an asymptote, approached from above as $x \to \infty$ and from below as $x \to -\infty$.

By the quotient rule,

$$g'(x) = \frac{2(1+x^2) - 2x \cdot 2x}{(1+x^2)^2} = \frac{2(1-x^2)}{(1+x^2)^2}.$$

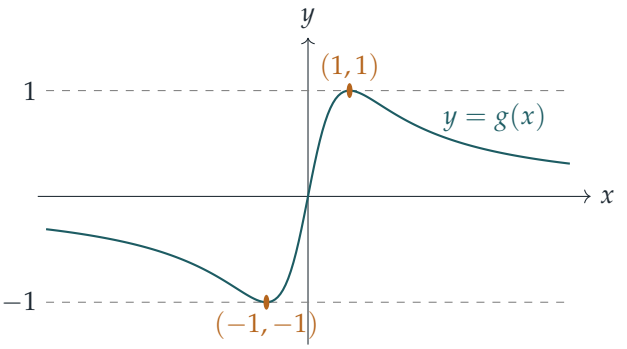
This is zero only at $x = \pm 1$. It is positive for $|x| < 1$ and negative for $|x| > 1$, so $x = 1$ gives a maximum, $g(1) = 1$, and $x = -1$ gives a minimum, $g(-1) = -1$. The gradient at the origin is $g'(0) = 2$.

We can confirm that these turning points are global without calculus:

$$1 - g(x) = \frac{(1-x)^2}{1+x^2} \geq 0, \qquad 1 + g(x) = \frac{(1+x)^2}{1+x^2} \geq 0,$$

so $-1 \leq g(x) \leq 1$ for every x, with equality only at $x = \mp 1$. Since g is continuous and takes the values -1 and 1, it takes every value in between (by the intermediate value theorem). Hence

$$\text{the range of } g \text{ is } \; -1 \leq g(x) \leq 1.$$



Part (ii)

By part (i), $\frac{2x}{1+x^2}$ always lies in $[-1, 1]$, the domain of $\sin^{-1}$, so f is defined for every real x.

Let $\theta = \tan^{-1} x$, so that $x = \tan\theta$ with $-\frac{1}{2}\pi < \theta < \frac{1}{2}\pi$. Then

$$\frac{2x}{1+x^2} = \frac{2\tan\theta}{1+\tan^2\theta} = \frac{2\tan\theta}{\sec^2\theta} = 2\sin\theta\cos\theta = \sin 2\theta,$$

and so

$$f(x) = \sin^{-1}(\sin 2\theta).$$

Ask yourself Isn't $\sin^{-1}(\sin 2\theta)$ just 2θ?

Only when 2θ is in the range of $\sin^{-1}$, namely $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$. By definition, $\sin^{-1} y$ is the *unique* angle in that interval whose sine is y. So to find $f(x)$ we must find the angle in $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$ with the same sine as 2θ.

Case $-1 \leq x \leq 1$**.** Since tan is increasing on $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$, this means $-\frac{1}{4}\pi \leq \theta \leq \frac{1}{4}\pi$, so $-\frac{1}{2}\pi \leq 2\theta \leq \frac{1}{2}\pi$. Then 2θ is itself the angle in the principal range with sine $\sin 2\theta$, so

$$f(x) = 2\theta = 2\tan^{-1} x \qquad (-1 \leq x \leq 1).$$

Case $x \geq 1$**.** Now $\frac{1}{4}\pi \leq \theta < \frac{1}{2}\pi$, so $\frac{1}{2}\pi \leq 2\theta < \pi$, which is outside the principal range. But

$$\sin(\pi - 2\theta) = \sin 2\theta \qquad \text{and} \qquad 0 < \pi - 2\theta \leq \tfrac{1}{2}\pi,$$

so $\pi - 2\theta$ is the angle in the principal range with the right sine. Hence

$$f(x) = \pi - 2\theta = \pi - 2\tan^{-1} x \qquad (x \geq 1).$$

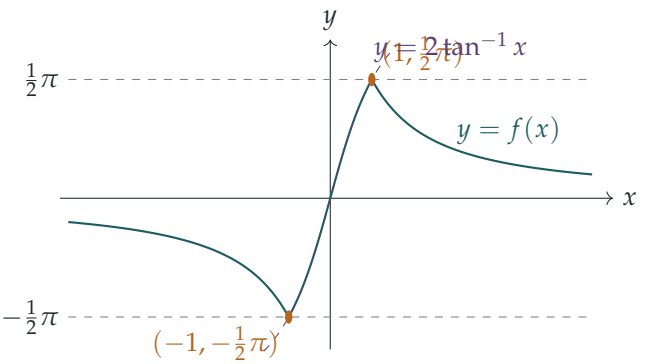
At $x = 1$ both formulae give $\frac{1}{2}\pi$, as they must. This completes the "show that".

For the sketch we also want $x \leq -1$. Since g is odd and $\sin^{-1}$ is odd, f is odd, so $f(x) = -f(-x) = -\pi - 2\tan^{-1} x$ for $x \leq -1$. (Equally: $-\pi < 2\theta \leq -\frac{1}{2}\pi$ there, and $-\pi - 2\theta$ is the angle in $[-\frac{1}{2}\pi, 0)$ with the same sine.)

Shape of the graph.

- On $[-1, 1]$, $f(x) = 2\tan^{-1} x$ increases from $-\frac{1}{2}\pi$ to $\frac{1}{2}\pi$, through the origin with gradient 2.
- For $x \geq 1$, $f(x) = \pi - 2\tan^{-1} x$ decreases from $\frac{1}{2}\pi$, and tends to $\pi - 2 \cdot \frac{1}{2}\pi = 0$ from above as $x \to \infty$. By oddness, for $x \leq -1$ it rises from $-\frac{1}{2}\pi$ towards 0 from below as $x \to -\infty$. So the x-axis is again an asymptote.
- The gradient is $\dfrac{2}{1+x^2}$ for $|x| < 1$ and $-\dfrac{2}{1+x^2}$ for $|x| > 1$. At $x = 1$ it jumps from $+1$ to -1 (and similarly at $x = -1$), so the graph has sharp corners at $(\pm 1, \pm\frac{1}{2}\pi)$: f is continuous but not differentiable there.

You can see the corners directly from the chain rule: $f'(x) = g'(x)/\sqrt{1 - g(x)^2}$, and part (i) gave $1 - g^2 = (1-x^2)^2/(1+x^2)^2$, whose square root is $|1-x^2|/(1+x^2)$. So $f'(x) = \dfrac{2(1-x^2)}{(1+x^2)\,|1-x^2|} = \pm\dfrac{2}{1+x^2}$, with the sign of $1 - x^2$.



The dashed curve shows $2\tan^{-1} x$ continuing past $x = 1$: it would climb above $\frac{1}{2}\pi$, which $\sin^{-1}$ can never do, so f "reflects" back down instead.

Checks

- At $x = \sqrt{3}$: $g(\sqrt{3}) = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$, so $f(\sqrt{3}) = \frac{1}{3}\pi$; and the formula gives $\pi - 2 \cdot \frac{1}{3}\pi = \frac{1}{3}\pi$. Agreed.

- At $x = 10$: $f(10) = \sin^{-1}(20/101) \approx 0.19934$, and $\pi - 2\tan^{-1} 10 \approx 0.19934$. A computer testing 10^5 random values of x in each of $[-1, 1]$, $[1, 10^4]$ and $[-10^4, -1]$ found agreement to 10^{-7}, and the one-sided gradients at $x = 1$ came out as $+1$ and -1.
- If f were $2\tan^{-1} x$ everywhere it would reach nearly π for large x, impossible for a value of $\sin^{-1}$; the reflection fixes this.

Common slips

- Writing $\sin^{-1}(\sin 2\theta) = 2\theta$ without checking that 2θ is in $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$.
- Drawing f with a smooth rounded top at $x = 1$. The graph has a corner there, because $\sqrt{1 - g^2}$ involves $|1 - x^2|$, not $1 - x^2$.
- Forgetting that f tends to 0 (not $\pm\pi$) as $x \to \pm\infty$.

Beyond STEP

Inverse trigonometric functions are a first example of choosing a *branch*: sin is not one-to-one, so we must restrict its domain before inverting, and identities like $\sin^{-1}(\sin\phi) = \phi$ then hold only on that branch. The same issue becomes central in complex analysis, where $\log z$ and $z^{1/2}$ need branch cuts. The corner at $x = 1$ is also a neat example of a function built from smooth pieces that is continuous but not differentiable at a point. For branches and multi-valued functions with plenty of pictures, see Needham, *Visual Complex Analysis*; for continuity and differentiability, Abbott, *Understanding Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.